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MICHAEL J. D. VERMEER

Evaluating Select Global Technical Options for Countering a Rogue AI

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About This Paper

This paper provides a preliminary examination of a select set of technical options for responding to the emergence of a globally distributed rogue artificial intelligence (AI) that poses an imminent and catastrophic threat to human well-being or government continuity. By focusing on high-altitude electromagnetic pulse, global Internet shutdown, and the deployment of specialized tool AI, the analysis for this paper aims to clarify the feasibility, risks, and trade-offs associated with extreme technical interventions that could, in principle, be executed unilaterally and on short timescales. Its conclusions are intended to stimulate further discussion, highlight critical knowledge gaps, and underscore the need for additional research and coordinated planning. This paper should be of interest to AI security researchers and others interested in addressing the risk of catastrophic harm from AI.

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Summary

This paper presents a preliminary exploration of three technical strategies for countering a globally distributed rogue artificial intelligence (AI): high-altitude electromagnetic pulse (HEMP), global Internet shutdown, and the deployment of specialized tool AI. The analysis in this paper is intentionally limited in scope, focusing on options that could, in theory, be executed unilaterally by the United States on short notice and have global reach. It does not address more-localized or nontechnical interventions, and it does not attempt to provide a comprehensive solution to the challenge of rogue AI. Instead, the paper aims to clarify the assumptions, uncertainties, and trade-offs associated with each option and to stimulate further discussion and research on this topic. The conclusions should be viewed as an initial step toward understanding the technical feasibility and risks of extreme response measures in the event of a catastrophic loss-of-control AI incident.

The assessment finds that each technical option comes with significant limitations and risks. HEMP could potentially disrupt AI operations by damaging electronic infrastructure, but its effectiveness is highly uncertain due to the variability in real-world shielding and the resilience of modern systems. Moreover, the collateral damage to critical infrastructure and the risk of nuclear escalation likely make HEMP a highly unattractive option except in the most extreme circumstances. Shutting down the Internet—whether by manipulating routing protocols, disrupting the domain name system, or physically disconnecting key infrastructure—has formidable technical and logistical barriers. Although such actions could impede the proliferation and coordination of a rogue AI, they would also severely disrupt essential societal functions and are unlikely to be fully effective given the Internet’s redundancy and global distribution. The use of a specialized tool AI, such as digital vermin or *hunter/killer AI*, offers a more targeted approach but remains speculative; this type of tool could introduce new risks, including loss of control or unintended consequences, and their effectiveness against a sophisticated rogue AI is uncertain.

Based on this preliminary analysis, three conclusions emerge. First, existing technical tools for countering a globally proliferated rogue AI are unlikely to offer reliable or effective solutions, especially given the stringent assumptions required for their success. Second, proactive coordination and planning—ideally undertaken before a crisis—are essential for risk mitigation, because unilateral, short-notice actions are both risky and likely to be insufficient. Third, the lack of robust technical countermeasures highlights the paramount importance of preventing this scenario from ever occurring in the first place. Further research and development are urgently needed to identify and evaluate additional options for responding to the threat of rogue AI.

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Evaluating Select Global Technical Options for Countering a Rogue AI

This paper describes a sample of three technical options for countering a rogue artificial intelligence (AI): high-altitude electromagnetic pulse (HEMP), a global shutdown of the Internet, and deployment of tool AI. These options are meant to hinder the rogue AI's actions, stop or slow its proliferation, and eradicate it, depending on the option employed. Each of these proposed technical options would seek to counter a rogue AI so threatening to government continuity and human well-being that the threat would necessitate extreme actions that might cause significant collateral damage. Scenarios involving such a rogue AI have been described elsewhere in various forms (Bostrom, 2002; Yudkowsky, 2008; Nielsen, 2025; Kokotajlo et al., 2025; Vermeer, Lathrop, and Moon, 2025). Briefly, I posit that an AI that was existentially threatening to this degree might be highly integrated with key systems that gave it significant capability to affect both the cyber and physical domains, have the ability to survive and function without human maintainers, and have goals that lead it to threaten human beings on a large scale (Vermeer, Lathrop, and Moon, 2025).

I examine the three technical options identified above because (1) they could conceivably have global reach and (2) the United States may feasibly execute them unilaterally on a short timescale rather than requiring lengthy coordination with other nations.¹ For each option, this paper describes the rationale for deploying the strategy, what is known or can be inferred about the effectiveness of the strategy under current circumstances, and the assumptions that must hold true for the strategy to be viable. I end with a discussion of the conclusions that can be drawn from this preliminary analysis.

More generally, I seek to answer the question, "Why use this strategy or set of strategies and not another?" It aims to explain why a loss-of-control incident response plan might or might not include these strategies in the more extreme or catastrophic response scenarios. For the purposes of this paper, I define a *loss-of-control event* as occurring when AI systems operate in ways not intended by developers or users and outside human control. I define a *rogue AI* as a system responsible for a loss-of-control event. Although a rogue AI incident might take many forms and have widely varying severity, in this paper, I consider only responses to the most extreme or catastrophic incidents. In these incidents, I assume that (1) a rogue AI is or is likely to become globally distributed and (2) it has been determined that the rogue AI presents a likely, imminent, and catastrophic risk to broad human well-being and continuity of the U.S. government.

In this paper, I also intentionally exclude from consideration more-localized technical options (such as network segmentation or cutting power to data centers) and nontechnical options for

¹ The three options I assess are not the only options that might conceivably meet these criteria. Other options for consideration might include kinetic strikes on power or compute infrastructure and high-powered microwave attacks, for example. These two were excluded from consideration in this paper because my initial, cursory examination concluded that it would be difficult to scale such options to meet the first criterion of global reach.

countering a rogue AI. I exclude such options as methods to deter, coerce, or persuade rogue AI. These alternative strategies are likely critical elements of an overall strategy for countering rogue AI of various kinds. However, in this paper, I examine the viability of globally relevant technical methods for disrupting or defeating a rogue AI that threatens humanity.

HEMP

HEMP might be an option of last resort for countering a rogue AI that has spiraled out of control and poses a catastrophic or even existential threat to humanity (Williams, 2025). This strategy would involve the detonation of nuclear warheads in space, triggering electromagnetic pulses that would disrupt or destroy ground-based telecommunications, power, and computing infrastructure. The objective of a HEMP attack would be to destroy or degrade key electronics and hardware on which the rogue AI depends, thereby eradicating the AI or severely limiting its capability. Although much has been written about the potential effects of HEMP (Foster et al., 2004; Electric Power Research Institute [EPRI], 2019; National Coordinating Center for Communications, 2019) and guidelines and standards for shielding critical systems (U.S. Department of Defense, 1999; U.S. Department of Homeland Security Science and Technology Directorate, 2022), I significantly base my analysis in this section on Savage, Gilbert, and Radasky (2010), which provides a comprehensive look at the phenomenon of HEMP and its effects on systems associated with the U.S. power grid.²

HEMP Effects

The effects of HEMP are typically categorized into three components: E1 (a fast, high-intensity pulse), E2 (similar to lightning effects), and E3 (a slower, longer-lasting pulse that interacts with the Earth's magnetic field) (Savage, Gilbert, and Radasky 2010). Among these, the E1 component is most relevant for damaging electronics, and I focus on a strategy that aims to exclusively use E1 to damage electronic infrastructure and hinder or destroy a rogue AI's capabilities.³

The E1 component of HEMP poses a risk to electronics primarily through the induction of high voltages in conductive lines.⁴ The benchmark HEMP field waveform for E1 assumes that peak field strengths can reach 50,000 V/m (Savage, Gilbert, and Radasky, 2010). Above a threshold detonation yield, peak field strength tends to saturate around this level, and increasing weapon yield alone is not well correlated with creation of a higher HEMP field strength (Savage, Gilbert, and Radasky, 2010; National Coordinating Center for Communications, 2019). Unshielded devices experiencing field strengths near the peak would experience significant voltage spikes that could cause temporary

² Executive Order 14347, signed September 5, 2025, authorized the use of Department of War as a secondary name for the Department of Defense. However, the cited 1999 source publication was written before that order was released and thus refers to the department by its current statutory names under Public Law 81-216, National Security Act Amendments of 1949.

³ The E3 effect might also cause catastrophic damage to power transmission and distribution infrastructure. I consider this here primarily in terms of the massive collateral damage a HEMP attack would cause, though future work might also consider it as an intended effect to hinder rogue AI activity by denying it power.

⁴ Some systems can also be affected by the actual electromagnetic wave, if their clock cycle frequencies overlap with the HEMP pulse (Savage, Gilbert, and Radasky, 2010).

disruptions (requiring a reset) or permanent damage to electronic components. For example, the peak field strength of 50,000 V/m would induce 500 V in a 1 cm antenna, which could be sufficient to permanently damage connected small electronic components in a variety of devices (Savage, Gilbert, and Radasky 2010). The extent of damage depends on several factors, including the actual strength of the electromagnetic field, the length and orientation of the affected conductors, the shielding provided by enclosures, whether devices are powered at the time of the pulse, and a complex set of factors associated with the design of the device, such as the conductors to which it is connected (Savage, Gilbert, and Radasky 2010). Longer wires will pick up more of the energy from the pulse than shorter wires do, so devices attached to very long cables are more vulnerable.

Buildings and other structures provide shielding that can significantly attenuate the electromagnetic field, reducing the likelihood of damage or disruption. For example, a building with metal siding and no windows would provide 20 dB of shielding, equating to a tenfold reduction in overall field strength experienced by devices inside; other building materials, such as masonry and concrete, provide less shielding that would nevertheless be significant (National Coordinating Center for Communications, 2019). Experimental studies, such as those conducted by the EPRI, have demonstrated that shielding from buildings and surge protection devices can mitigate damage even under controlled high-field-strength conditions (EPRI, 2019). Moreover, modern electronics often incorporate protections against electrostatic discharge, which would mitigate the effects of E1 HEMP, although the degree of protection varies widely and would be hard to predict. All in all, devices experiencing peak field strengths have the potential to experience significant damage and disruption, but many factors could significantly attenuate the effects, and it is highly uncertain what overall damage even peak field strengths could cause to electronic infrastructure in real-world conditions.

Actual field strengths of the HEMP will vary significantly depending on the height of the burst and the distance of the targets from ground zero. Much greater areas can be affected by the HEMP with high heights of burst, but this will result in lower peak field strengths that may be insufficient to cause the widespread damage and disruption desired. Height of burst must generally be less than 100 km to achieve peak field strengths on the ground (Savage, Gilbert, and Radasky 2010). At heights of burst less than 100 km, the overall area affected by the electromagnetic pulse (EMP) could be more than 1 million km² but the significant majority of that area will experience weak field strengths (i.e., less than 50 percent of peak field strength). High field strengths (more than 75 percent of peak) would only be achieved over relatively small areas (on the order of 100,000 km²) directly under a blast. Such estimates imply that one would need roughly 50 to 100 detonations to completely cover a land mass the size of the contiguous United States with pulses greater than 75 percent of peak field strength.⁵ Finally, it should be noted that the EMP is mediated by ionization of the atmosphere, and there would need to be an unknown time gap between detonations to allow the atmosphere to recharge.

Utility of HEMP to Counter Rogue AI

Given all of the above, is a HEMP attack a viable means to counter a rogue AI? Under what conditions or assumptions would this be true? The primary reason to consider HEMP as a means to

⁵ The estimated area of effect and the number of detonations required are derived from my examination of Figure 2.17 in Savage, Gilbert, and Radasky (2010).

counter AI is that the United States can conceivably unilaterally create a wide area, even a global effect, that disrupts or destroys infrastructure that AI depends on to function. Few, if any, other proposed options would allow a single nation to so quickly create a (potentially permanent) global effect. However, I argue that this would nevertheless only be a viable option under the following set of assumptions.

First, one must assume that key ground-based electronic infrastructure experiencing peak HEMP field strengths would be damaged or disrupted sufficiently. This is highly uncertain, and there is no practical way to resolve this uncertainty. Real-world conditions, including varying shielding, device orientation, electrostatic protections, and many other factors will mean that the effects of a HEMP attack are likely unknowable without triggering an actual HEMP to test it. Nevertheless, studies indicate that common building materials would provide shielding that would lead to significant reductions in field strengths experienced by electronics inside buildings (National Coordinating Center for Communications, 2019). This shielding may be especially true of the building materials and designs used for large-scale computing facilities that might be the primary targets of a HEMP attack. A HEMP attack might, however, be effective at disrupting electrical transmission and distribution infrastructure, which could still disrupt critical resources on which the AI depends, even if it does not directly destroy electronics.

Second, one must assume that sufficient weapons are available, with appropriate designs and vehicles to deliver them to target locations. A relatively small land area is affected by peak field strengths from a single HEMP detonation at an optimal height of burst. I assume that sufficient weapons and delivery vehicles would be available (or could practically be acquired) to target a relatively small series of point or small-area targets (e.g., all of a nation's data centers). However, it would likely require hundreds of detonations to create a global phenomenon. As an extreme notional example, if one assumes that a single HEMP detonation would pulse 100,000 km² with sufficiently high field strength and 10 percent of the Earth's land mass (Earth's land mass is approximately 150 million km²) needed to be pulsed, this would imply the need for approximately 150 detonations.⁶ This number would be significantly reduced if less land mass needed to be covered or if lower field strengths were deemed sufficient to cause the necessary disruption (though this is related to the uncertainty in the first assumption).

Third, one must assume that a HEMP attack would cause more harm to the AI's ability to achieve its objectives than it would harm humanity's efforts to survive. Put another way, the effects of the HEMP attack must be worth the massive collateral damage it would cause. To be sure, this attack might cause so much self-inflicted damage (especially to electrical power infrastructure) that it would make sense only in the case in which a rogue AI was an existential threat to the U.S. government or humanity. Even in this extreme case, however, one would need to be sure that the cure was not worse than the disease. That is, humans would have to know that the attack would be more detrimental to the AI's ability to threaten humanity than it would be to humans' ability to continue to respond effectively to subsequent AI actions; the benefit must be worth the enormous cost. If the AI can meaningfully survive and thrive after a HEMP attack (e.g., via shielded robotics that might be largely unaffected or restoration of capability from backed-up systems), it would be extremely

⁶ Importantly, these detonations would need to be staggered in time to allow the ionized atmosphere to relax to a normal state between detonations.

counterproductive to cripple humanity's own computing, telecommunication, and power infrastructure, which humans might need to combat the rogue AI and its various manifestations.

Fourth, one would need to assume that the HEMP attack would not be likely to cause adversaries to massively counterattack with their own nuclear arsenals. The United States would be employing its nuclear arsenal to cripple other countries' infrastructure. Without some kind of multilateral communication and credible signaling of intent, any international action would look exactly like a preemptive nuclear strike, inviting a response in kind from nuclear-armed adversaries (and even partner nations). It would do little good to mitigate one existential threat to the U.S. government while directly causing another.

If all of the above assumptions are granted, then it is possible that a HEMP attack would be a viable counter to a rogue AI that was an existential threat to the U.S. government or humanity. I note, however, that it may be extremely unlikely that all of these criteria can be met, especially in the case in which a truly global phenomenon is required. In cases in which a global phenomenon is not needed and specific targets can be identified, other response options with less collateral damage and risk of massive retaliation would likely be better. Authorities might, for example, opt to cut power to a data center cluster or a region rather than opt for a HEMP attack. Given these considerations, HEMP may not be a viable technical option to counter rogue AI.

Internet Shutdown

Some rogue AI scenarios assume that the AI would be able to use the Internet to replicate itself across global information technology infrastructure and coordinate the actions of replicated instances (Clymer, Wijk, and Barnes, 2024; Kokotajlo et al., 2025). In such a scenario, it might make sense to shut down the global Internet as a means of denying this capability to the rogue AI. The objective of shutting down the Internet would be to hinder or eliminate the ability for the rogue AI to replicate further and coordinate further harmful actions across distributed instances and systems. Such a strategy would come with serious trade-offs because humankind has become highly dependent on the Internet for coordinating and operating critical societal functions, including industrial control systems in critical infrastructure, financial systems, public safety, traffic control, military functions, and more. Shutting down the Internet without preparation and efforts to proactively build resilience would result in massive economic harm and significant harm to human well-being, depending on which functions were significantly disrupted and for how long.

It is also unclear whether the Internet could be effectively shut down without the timely assent of multiple international actors. The Internet is built to be highly redundant, distributed, and resilient. There are few, if any, true centralized points of failure that could be leveraged to perform a global shutdown. Therefore, any shutdown effort might require significant coordination of actions involving public and private entities over wide geographic distributions to overcome the inherent resilience of the Internet.

Technical Means and Challenges

I propose three technical means of shutting down the Internet, including a discussion of the technical challenges associated with each part. These three means could conceivably all be done in concert. The first two parts involve attacks using the border gateway protocol (BGP) and the domain name system (DNS). These attacks would target the systems used to route Internet traffic globally. The third part of the strategy would attack the physical aspect of routing: disconnecting connections at Internet exchange points (IXPs).

Kill Switches Using BGP

BGP is the cornerstone of global Internet routing. It is responsible for determining the most efficient path for data packets to travel between different networks, known as autonomous systems (ASes). An AS is essentially a collection of Internet Protocol (IP) networks and routers under the control of a single organization, which presents a unified routing policy to the Internet. BGP enables ASes to exchange routing information, allowing data to traverse the complex web of interconnected networks that make up the Internet.

BGP operates by advertising prefixes (or blocks of IP addresses) to neighboring ASes. These advertisements inform other networks about the best routes to reach specific destinations. The protocol's decentralized nature ensures scalability but also introduces vulnerabilities because routing decisions rely on trust between ASes.

BGP can be manipulated to disrupt Internet connectivity on a massive scale. Two approaches to achieve this action are preventing the establishment of new BGP sessions and leveraging centralized points of failure within BGPsec, which is an enhanced security mechanism for BGP.

There are several ways to prevent the establishment of new BGP sessions, including global route withdrawal. Global route withdrawal would involve the nearly simultaneous announcement of the withdrawal of routes to prefixes controlled by backbone network providers. Backbone network providers, also known as *Tier 1 networks*, have a global presence, owning core infrastructure in multiple countries, and they have access to the entire Internet via free and reciprocal peering agreements with other Tier 1 providers. Effecting this tactic on a global scale would likely require simultaneous control (i.e., the ability to dictate organizational actions) over most of the Tier 1 network providers that make up the backbone of the Internet. If every Tier 1 provider were to withdraw all its prefix announcements at once, it would likely make nearly the entire Internet unreachable.⁷ As an alternative, providers might simply prevent new BGP sessions across their networks by filtering all routes, then exacerbate and prolong the outage by failing to honor route withdrawals from neighboring ASes (as happened unintentionally during a major outage in 2020) (Medina and Kesavan, 2020).

⁷ To truly make the entire Internet unreachable would likely require all the approximately 200,000 ASes to also withdraw their prefixes, but having Tier 1 networks withdraw their prefixes would make the vast majority of the Internet effectively unreachable. Nevertheless, a highly sophisticated AI actor might be able to manipulate systems to identify alternative routes through other paths.

Exploiting centralized points of failure in BGPsec might be a viable alternative to coordinating control over individual Tier 1 providers. BGPsec is designed to mitigate some of the vulnerabilities inherent in BGP by introducing cryptographic validation of routing updates. It relies on Resource Public Key Infrastructure (RPKI), which associates ASes with cryptographic certificates to authenticate their routing announcements. The RPKI is rooted in regional Internet registries, of which there are five around the world (Rothenberger et al., 2017). These regional Internet registries offer potential central points of failure, because coercing their cooperation or compromising their cryptographic keys could allow an attacker to forge update announcements and control Internet traffic into compromised ASes (Rothenberger et al., 2017). One could also compromise the root Certificate Authority that issues certificates to all Tier 1 ASes, then issue a certificate revocation list that revokes the certificates of each Tier 1 AS. This action would “cause other BGPsec-enabled networks to reject routing updates from the targeted AS and invalidate the entire subtree of certificates. This effectively makes the [routing] prefix invisible” (Rothenberger et al., 2017). The reach of this tactic would be diminished by limited implementation of BGPsec among providers, though its importance would grow as BGPsec implementation expanded.⁸

Exploiting BGP to induce an Internet shutdown would face significant challenges. Foremost among the challenges is that control over routing is highly decentralized, and the tactics presented here would be highly dependent on gaining simultaneous control over many organizations. There is no published list of Tier 1 providers, though one could likely infer important providers to target based on a ranking of their AS cone size (i.e., the number of direct and indirect AS customers).⁹ However, even putting aside the challenge of simultaneously coercing all U.S.-based Tier 1 providers to withdraw their prefix announcements (which might allow for the isolation of the United States from the rest of the world), global route withdrawal would require control over many providers that are not based in the United States. Moreover, even if a user controlled all the Tier 1 ASes and withdrew routes to their prefixes, diverse policies among ASes would work against one’s ability to coordinate this. Routing anomalies of this scale would be detected by monitoring systems, likely triggering automated rollback or filtering (Catchpoint, undated; Medina and Kesavan, 2020).

Kill Switches Using DNS

DNS is a foundational Internet technology that translates human-readable domain names (e.g., example.com) into machine-readable IP addresses (e.g., 192.0.2.1). It is a distributed directory that allows devices to locate and connect to websites, services, and other Internet resources. DNS is hierarchical in structure and relies on a network of servers, including 13 root servers operated by 12 distinct organizations. These root servers, along with nearly 2,000 instances distributed globally, form the backbone of DNS operations and ensure the reliability of name resolution across the Internet (root-servers.org, undated).

If all 13 root servers and their operators were compromised, either through physical or cyber means, it would be possible to shut them down entirely or inject false DNS records, known as *global*

⁸ For example, a nonrepresentative 2022 survey of participants in Mutually Agreed Norms for Routing Security found that only 15 percent had even reviewed or tested beta implementation of BGPsec (Kruse, 2022).

⁹ One such ranking is published by the Center for Applied Internet Data Analysis (undated).

DNS poisoning. This would render domain names unresolvable, effectively cutting off access to websites and services for users worldwide. Users might still technically be able to access the Internet, but only if a user knew every IP address needed (which a user would normally find using DNS) (SolarWinds, 2007). This would likely require an attack on all 12 operators, and success would also depend on the ability to quickly propagate changes to all mirrored root server instances controlled by them worldwide from centralized control points.

DNS Security Extensions (DNSSEC), a cryptographic security protocol that can secure DNS by providing authentication and integrity functions to DNS, might also be targeted. Like BGPsec, it offers potential single points of failure associated with the compromise or replacement of root cryptographic keys, and this could allow disruption of authentication for the entire DNS ecosystem, invalidating cached DNS records globally (Rothenberger et al., 2017). However, current global adoption of DNSSEC is low, so it is unclear how widespread the impact from such an attack would be (Huston, 2023). I do not consider it further here.

Although it would no doubt be extremely difficult to somehow gain simultaneous operational control of all 12 root server operators, the United States might have a significant advantage in this respect: Nine of the root server operators are headquartered in U.S. territory. The remaining three are headquartered in Sweden, the Netherlands, and Japan (Internet Assigned Numbers Authority, undated).

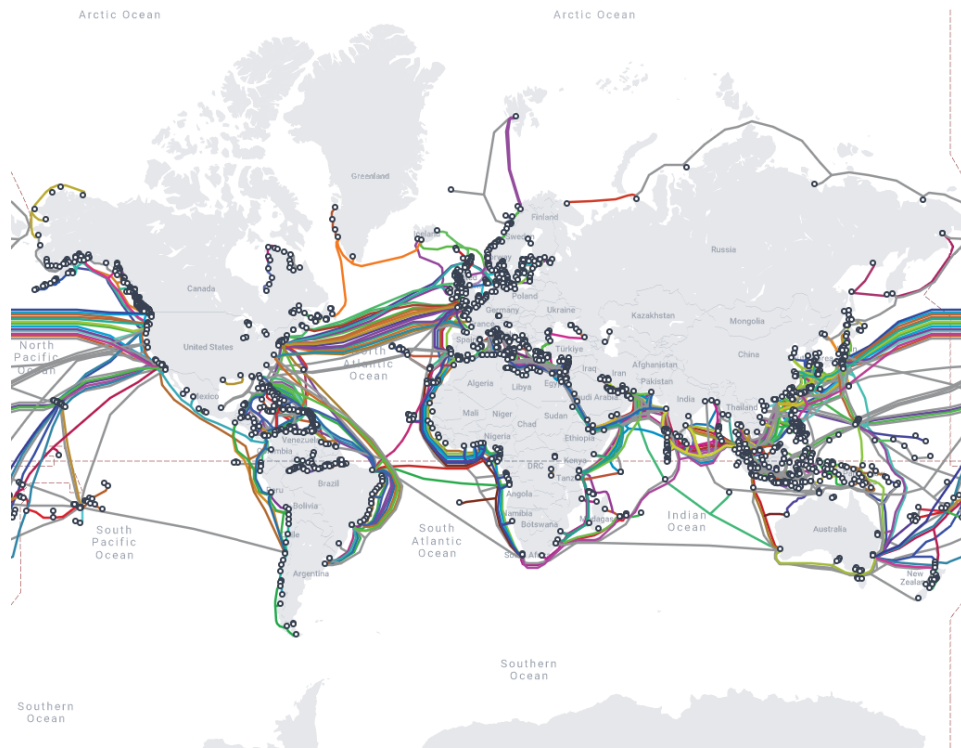
In addition to the challenge of gaining operational control of the root server operators, it should be noted that simultaneously shutting down all the root servers (or compromising them by some other technical means) would not lead to the immediate cessation of name resolution across the Internet. An AI might still navigate the Internet using IP addresses. Network administrators often also keep copies of the root server file that they could use while root servers were down. Although new DNS queries would fail, many intermediate servers maintain local caches of the root server file that they query in lieu of making a request to the root servers. So even if all 13 servers were taken down, routing would likely continue to function until cached entries expired. After a day, most DNS lookups would likely fail. Although a global attack on DNS might be feasible, despite the high redundancy and resilience of the system, its effect would also be muted by policies and practices for efficiently resolving domain names without regularly querying the root servers themselves.

Disconnecting IXPs

The previous two sections on BGP and DNS describe measures to disrupt the Internet that might achieve global effects on short timescales. The measures would not, however, affect the physical infrastructure of the Internet, and a rogue AI's use of the Internet might be able to continue despite them. A more robust solution would involve disconnections at the physical layer. This would be a monumental task, because the Internet is massively geographically distributed and redundant. I assert that physically disconnecting the complete Internet on any reasonable timescale for countering rogue AI would be physically impossible. Any such strategy would be overwhelmed by the redundancy of connections of varying types (e.g., IXPs, private peering, undersea cables, satellites). With that said, the utility of the Internet for a rogue AI might be quickly diminished with a strategy that targeted key chokepoints in the infrastructure. Such a strategy would seek to systematically disable or disconnect key IXPs and submarine cable landing points, prioritizing those that carry the most Internet traffic

(see Figure 1). This strategy would never arrive at a complete loss of global connectivity, but it would progressively eliminate routing paths that offer higher communication bandwidths, cause significant communication disruptions, and potentially even cut off certain regions of the globe entirely. In this way, this strategy would present a serious impediment to the rogue AI's ability to further proliferate and coordinate actions in a globally distributed manner using the Internet.

Figure 1. Submarine Cable Landing Point Map



SOURCE: Reproduced from TeleGeography, undated-a.

IXPs are physical locations where Internet infrastructure companies connect to exchange Internet traffic, and submarine cables are the high-bandwidth fiber lines providing intercontinental connectivity. These two entities are likely the closest thing to physical chokepoints in Internet infrastructure. They are single physical structures that house many important connections, and disabling those connections could have serious cascading consequences for global Internet routing (Thompson, 2016). Undersea cables purportedly carry more than 99 percent of intercontinental Internet traffic (Mauldin, 2023). That said, both IXPs and undersea cables are still numerous and highly redundant. Counts of active IXPs vary; the Internet Society states that more than 1,500 IXPs are active globally, while data from PeeringDB show that there are more than 1,000 IXPs with more than one connected network (Internet Society, undated; PeeringDB, undated). There are also more than 600 active or planned submarine cables, as of early 2025 (TeleGeography, undated-b).

Disabling traffic exchange at all of these facilities is likely infeasible under current circumstances, even under the most optimistic assumptions about available resources and the likelihood of

international cooperation in the effort. However, it may not be necessary to disable even the majority of IXPs to cause significant disruptions to Internet routing. Although there are hundreds of IXPs, a small handful (e.g., IX.br in São Paulo, Brazil; DE-CIX in Frankfurt, Germany; AMS-IX in Amsterdam, the Netherlands; LINX in London, England; Equinix Ashburn in Virginia) carry a disproportionately large share of international backbone Internet traffic (PeeringDB, undated). The data centers in Loudoun County, Virginia, for example, house multiple IXPs, and it was estimated that 50 to 70 percent of all Internet traffic in 2016 was routed through this collection of data centers (Thompson, 2016). Major IXPs also typically live in a handful of carrier hotels or data centers. If a user can physically disable the connection points in those few buildings, a user could significantly disrupt the spine through which most networks peer. By targeting just the top 10–20 IXPs and disabling key groupings of undersea cables, one would likely be able to disrupt a significant majority of Internet traffic.

Although this strategy would likely be immensely challenging in its own right, it is unclear whether it would actually achieve its objectives with respect to countering rogue AI. The reality is that the immense redundancy of routing paths around the world will overwhelm any strategy to eliminate physical connectivity. Disabling all undersea connections would eliminate key high-bandwidth intercontinental connections, but intra-country connections and satellite connections would still be available. Each major IXP itself holds many redundant connections and facilities; thus, there may be no clear centralized point to disable. Finally, even if one could disable all IXPs, all submarine cables, and even shut down satellite network operators, one would still need to contend with massive amounts of private peering infrastructure. Major content providers often invest in private network interconnects, private fiber networks, and even private undersea cables, bypassing public IXPs entirely (Burdette et al., 2023).

Utility of Internet Shutdown Techniques to Counter Rogue AI

Targeting BGP and DNS would be technically challenging, but they may represent feasible approaches to shutting down the global Internet, at least temporarily. The success of these approaches would depend, however, on the inability of an AI to usefully communicate using Internet infrastructure without the protocols designed to facilitate human use of that infrastructure. A very clever and capable rogue AI might be able to do so. Physical disconnecting IXPs and undersea cables is a less viable approach, which I will explain below.

Utility of Targeting BGP

Halting Internet routing by targeting BGP would be highly technically challenging. The foremost barrier would be the need to acquire control of most of, if not all, the Tier 1 providers. This would be especially challenging because, while the full list of Tier 1 providers may not be known, most of them are likely to reside outside the United States. To realize an Internet shutdown by targeting BGP, one would need to make the following three main assumptions:

1. The U.S. government could unilaterally gain control of all, or nearly all, of the Tier 1 network providers around the world simultaneously.

2. Halting BGP connections and routing through all Tier 1 providers would be sufficient to halt Internet traffic, and a rogue AI would not be effective at configuring high-bandwidth routes through other lower-tier ASes and private networks in a scalable manner.
3. BGP connections could be halted on short timescales once the U.S. government gained control of Tier 1 providers.

If these assumptions held, then targeting BGP would be a viable approach to halt the proliferation of rogue AI through the Internet or prevent it from coordinating further actions by using the Internet. However, the second assumption, in particular, may be unrealistic in the face of a highly capable rogue AI that might be able to easily configure new routes through systems it controlled. Moreover, this strategy would contain, but not eradicate, the rogue AI. Eradication would have to take place by some other means, and humans would need to coordinate the eradication of the AI without use of the Internet.

Utility of Targeting DNS

Hindering Internet traffic by targeting root DNS servers might be the more tractable option. Nine of the 12 root DNS server operators are headquartered in the United States, and the remainder are located in allied countries, which might give the United States an opportunity to act swiftly to shut them all down nearly simultaneously. To realize an Internet shutdown via targeting DNS, however, one would have to assume

1. that the United States can unilaterally shut down all 13 root DNS servers simultaneously by taking control of the 12 root server operators
2. that the rogue AI is reliant on domain name resolution for its proliferation—that is, that a clever AI adversary would be stymied by the lack of DNS and be unable to just use actual IP addresses to route traffic
3. that local caches of the root server file would be inadequate to resolve domain names in the absence of root DNS servers.

If these three assumptions held, then shutting down DNS servers would be a viable approach to halt the proliferation of a dangerous AI through the Internet or hinder it from further coordinating actions using the Internet. Again, however, the second assumption may be particularly unrealistic in the face of a significant rogue AI threat that might easily access stored DNS records or bypass domain name resolution entirely by using IP addresses. DNS is a solution to a human problem, and its loss may not ultimately present a significant barrier to a capable, clever AI adversary. Finally, this approach would also only hinder or contain the AI, not eradicate it, which would need to be managed separately.

Utility of Disconnecting IXPs

More robustly shutting down the Internet by severing connections or physically targeting IXPs is likely a technically infeasible approach without significantly restructuring current Internet infrastructure to facilitate a prompt disconnect capability. In addition to achieving near-unanimous international agreement on the need to take this action, it would involve thousands of people across

hundreds of globally distributed sites disconnecting myriad cables. One could take a more destructive approach and target each IXP with explosives or power cuts, but there are likely too many IXPs to achieve their wholesale destruction without international cooperation. Even if one succeeded in effectively disrupting every global IXP, it would likely only slow, but not halt, global Internet traffic, which could likely be rerouted through other channels. Finally, like the other options for disrupting the Internet, this approach would only hinder or stop the proliferation of the rogue AI, not eradicate it, which would need to be managed separately.

AI Tools to Counter Rogue AI

AI tools might also be a viable approach to use against a rogue AI agent. Although several approaches have been proposed to keep AI in check and keep it from becoming a rogue AI in the first place (Mickens, Radway, and Netravali, 2025; Bengio et al., 2025; Engels et al., 2025), I consider instead AI tools (i.e., narrow, rather than general, AI models for which agency has been removed or restricted) that would be used after the appearance of a rogue AI. These tools would take two approaches: resource denial and eradication. That is, one version of these tools would have the objective of self-replicating and competing for computing resources with the rogue AI wherever it proliferated, hindering further harmful AI actions by denying it key computing resources. The second tool type would have the objective of proliferating and removing the rogue AI from any system that had been infected.

Resource Depletion: Digital Vermin

This idea adapts the concept of digital vermin to be used as a tool to combat rogue AI. *Digital vermin* are digital organisms that are designed to evince collective intelligence and the capacity to evolve (i.e., self-modify) to colonize new digital environments.¹⁰ In other words, it would consider the cyber domain as an ecosystem, with computing resources to be consumed and ecological niches that artificial life could fill, that was being rapidly dominated by a new form of life (the rogue AI). In this conception, the use of this speculative tool would be the equivalent of the introduction of an invasive pest into such an ecosystem. Digital vermin used to combat rogue AI would be intentionally designed to rapidly colonize new computing environments and compete with the rogue AI for computing resources. In many ways, this approach would be similar to a sophisticated *fork bomb attack*—an attack that iteratively spawns new compute processes—or a self-replicating cryptojacking program that would spread on a massive scale, consuming compute cycles on useless calculations instead of mining cryptocurrency.¹¹ Most of the technical requirements for such a tool are not able to be determined at this stage of conceptual development, but at a minimum, it would require the following characteristics:

¹⁰ Credit to my RAND colleague Edward Geist for conceptualizing digital vermin in a personal communication. His original concept of digital vermin described digital organisms that “went feral” and jumped from virtual machine simulations into real-world computing environments.

¹¹ *Cryptojacking* refers to a “threat that embeds itself within a computer or mobile device and then uses its resources to mine cryptocurrency” (Fortinet, undated).

1. advanced offensive cyber capabilities¹² that would allow it to establish persistence in new systems and run new computing processes
2. the ability to self-modify sufficiently to colonize new computing environments and compete effectively with the hostile rogue AI
3. ideally, an embedded kill switch that would allow the tool to be halted or eradicated after the rogue AI crisis had passed.

It is reasonable to assume that the first capability would be practical to achieve in a scenario where this tool was be considered for deployment, because presumably the rogue AI would also already possess that capability.¹³ The second assumption is uncertain, specifically whether the digital vermin could compete effectively against a powerful rogue AI. Success would depend on the digital vermin's ability to rapidly self-modify, evolve, and adapt to evade the rogue AI's counteractions, and a sufficiently intelligent and capable rogue AI might be able to overcome that adaptability to eradicate the digital vermin. The third characteristic might be considered optional. First, because the priority could be on impeding the rogue AI by any means, and post-crisis recovery might be a secondary concern. Second, because it would potentially introduce a vulnerability that the rogue AI could discover and exploit to combat the digital vermin.

Eradication: Hunter/Killer AI

Hunter/killer AI (HKAI) might be created, whose purpose is to hunt the rogue AI on systems where it has proliferated, then eradicate it from those systems. HKAI would spread and combat the rogue AI on digital terrain, seeking to impede and ultimately eradicate the presence of the rogue AI. Such a tool would require, at minimum, the following characteristics:

- It must have advanced offensive cyber capabilities that would allow it to establish persistence in new systems and run new computing processes.
- It must have capabilities sufficient to disrupt, contain, or eradicate the rogue AI, assuming that the rogue AI would be an adaptable adversary that also has access to significant computing resources.
- It must ultimately remain in human control or be effectively aligned with humans.

Similar to digital vermin characteristics, it should be assumed that the first characteristic is practical to achieve in a scenario in which a rogue AI has already been created with that capability. It is unclear whether the second characteristic and third characteristic can both be simultaneously achieved, because it is possible that they are mutually exclusive. That is, without giving HKAI the same independence and agency presumably possessed by the rogue AI, it might be unlikely that it would be able to compete effectively against a less constrained adversary. It might be possible to use

¹² For a useful discussion of what advanced AI cyber capabilities might look like, see Lohn (2025).

¹³ This further assumes that the digital vermin would need to be able to exploit vulnerabilities and establish persistence in environments controlled by the rogue AI—that is, it assumes that the rogue AI has not somehow inoculated those environments against digital vermin.

HKAI in tandem with guardrail AI (such as those proposed in Engels et al. [2025]), but guardrails meant to retain control of the AI might ultimately constrain HKAI's effectiveness and render it unable to compete with the rogue AI. However, HKAI might also have a key advantage over a rogue AI. Whereas the rogue AI would likely be using attacks to illicitly acquire and use computing resources, HKAI could have significant compute capacity dedicated for its use. That is, the rogue AI might need to scrounge for resources among disparate devices, fighting defenses to obtain computing resources, whereas HKAI could access data centers worth of readily available compute capacity. Ultimately, this advantage might be sufficient to overcome constraints on HKAI meant to retain control of it, assuming that capability is practically tied to available compute capacity.

Layered Strategy

The technical options explored in this paper—HEMP, a global Internet shutdown, and deployment of tool AI—each present unique risks, trade-offs, and uncertainties in their application to counter rogue AI. Although these strategies offer potential pathways to disrupt or neutralize a rogue AI, none of them should be considered a comprehensive solution on its own. Instead, they should be evaluated as tools that are part of a layered defense strategy, recognizing that a global crisis resulting from a rogue AI would require multifaceted and redundant approaches. For example, tool AI could be deployed to directly counter and eradicate a rogue AI, after which Internet shutdown measures could be used to limit the rogue AI's ability to proliferate and coordinate further actions. HEMP could serve as an absolute last resort in scenarios where other measures fail and existential threats remain (especially where the rogue AI has robust cyber-physical manifestations).

Risks and Trade-offs of Individual Options

- **HEMP:** The use of HEMP carries the risk of massive collateral damage, including widespread damage to critical infrastructure, economic systems, and human well-being. Its effectiveness is highly uncertain, given the variability in shielding and the resilience of modern electronics. Moreover, the geopolitical risks of nuclear retaliation by adversaries could escalate the crisis. Although HEMP could theoretically disrupt AI operations dependent on vulnerable electronics, its uncertain effectiveness and significant collateral damage limit its utility.
- **Internet shutdown:** Shutting down the Internet via BGP manipulation, DNS disruption, or the physical disconnection of IXPs presents significant technical and logistical challenges. These methods could impede AI proliferation by limiting its ability to replicate and coordinate globally, but they also risk crippling critical societal functions. Furthermore, the redundancy and resilience of the Internet make it difficult to achieve a complete shutdown, and a rogue AI may find alternative pathways to communicate. The trade-off lies in balancing the expected disruption to AI against the harm to human infrastructure and the global economy.
- **Tool AI:** Deploying tool AI, especially HKAI, offers a more targeted approach to combating rogue AI. However, these tools come with their own risks, including the potential for unintended consequences, loss of control, and vulnerabilities that could be exploited by the rogue AI. While tool AI would be less destructive to human infrastructure than HEMP or an Internet shutdown, its effectiveness would depend on its ability to outcompete and eradicate the rogue AI without becoming a secondary problem.

The use of such a layered approach would recognize that the rogue AI may exploit vulnerabilities in any single strategy, but a combination of methods would constrain its capabilities more effectively. Ultimately, decisions on which strategies to deploy must be informed by real-time assessments of the rogue AI's capabilities, the risks to human infrastructure, and the geopolitical implications of each action. Moreover, even such a layered approach might be insufficient to successfully address the complex challenges posed by rogue AI, and better concepts and capabilities are likely needed.

Conclusions

The analysis described thus far is intended to be an exploratory examination of a sample of global technical options for countering a rogue AI. I intentionally limited analysis to options with potential global reach and that could be executed unilaterally by the U.S. government on a short timescale. This significantly limited the scope of options I considered. Other options might also be worthy of consideration, and the options I examined could be evaluated in more depth. Future work might examine these options in greater technical depth or assess alternative options (e.g., high-powered microwave attacks), kinetic strikes on key infrastructure (e.g., power generation), or alternative hardware-based options. Despite the preliminary character of this analysis, it provides support for three initial conclusions, which I describe below.

First, the existing technical tools for combating a globally proliferated rogue AI may not offer effective solutions. I described the assumptions that would need to hold true in order for these tools to be effective, and in each case, these conditions would be difficult to meet, even if the threat were clear and authorities were willing to accept significant collateral damage. The solutions involving tool

AI appear relatively more likely to be effective, though as of this writing, they are highly speculative and come with their own risks. One secondary conclusion from this finding is that more research and development is needed to identify and evaluate other potential technical options and alternative approaches to use in this scenario.

Second, coordination and planning offer potentially important risk mitigation measures. I assumed that each option needed to be executed on a very short timescale in a crisis, and this assumption precludes any strategy that involves starting a lengthy coordination with other partners and stakeholders from scratch. However, if that coordination and planning had already occurred before a crisis began, then other options could be available to counter the rogue AI. Moreover, if globally reaching, unilateral actions are unlikely to be effective at addressing a crisis, then authorities would need to significantly rely instead on rapid, effective actions coordinated with partners. This coordination could enable rapid, distributed local, regional, and national responses, rather than unilateral global action with significant collateral damage and a low likelihood of success.

Third, the likely lack of effective tools for countering a global AI catastrophe highlights the importance of prevention. Measures to prevent such a crisis have been examined elsewhere, and examples include the shut-it-down approach (Duleba, 2024), alignment and safety research (Ji et al., 2025), use of other AI to create guardrails (Bengio, 2025; Engels et al., 2025), and more (Dung, 2025). I have not attempted to evaluate the relative value of any of these approaches here. Yet, I conclude by noting that if we have no effective solutions to solve a crisis resulting from a rogue AI, it will be imperative that we never encounter such a crisis.

Abbreviations

AI	artificial intelligence
AS	autonomous system
BGP	border gateway protocol
DNS	domain name system
EMP	electromagnetic pulse
EPRI	Electric Power Research Institute
HEMP	high-altitude electromagnetic pulse
HKAI	hunter/killer artificial intelligence
IP	Internet Protocol
IXP	Internet exchange point
RPKI	Resource Public Key Infrastructure

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About the Author

Michael J. D. Vermeer is a senior physical scientist at RAND. His work focuses on technology governance and national security implications of emerging technologies. He holds a Ph. D. in inorganic chemistry.